Exercise Series 10 (solutions)

Exercise 1 (a) $X_1, \dots, X_n \overset{iid}{\sim} \text{Bernoulli}(p)$

$$\Rightarrow \sum_{i=1}^n X_i \sim \text{Binomial}(n,p)$$

Power function: $\beta_{left}(p) = \mathbb{P}_p\left(\sum_{i=1}^n X_i < c_{left}\right)$

$$= \mathbb{P}_p\left(\text{Bin}(n,p) < c_{left}\right)$$

$$= \text{CDF of } \text{Bin}(n,p) \text{ at } \lceil c_{left} - 1 \rceil \leftarrow \text{ceiling}$$

$$= \sum_{t=0}^{\lceil c_{left}-1\rceil} \binom{n}{t} p^t (1-p)^{n-t}$$

$$\beta_{right}(p) = \mathbb{P}_p\left(\sum_{i=1}^n X_i > c_{right}\right) = \mathbb{P}_p\left(\text{Bin}(n,p) > c_{right}\right)$$

$$= 1 - \mathbb{P}_p\left(\text{Bin}(n,p) \le c_{right}\right)$$

$$= 1 - \text{CDF of Bin}(n,p) \text{ at } c_{right}$$

$$= 1 - \sum_{t=0}^{\lfloor c_{right}\rfloor} \binom{n}{t} p^t (1-p)^{n-t}$$

$$= \sum_{t=\lfloor c_{right}\rfloor+1}^{n} \binom{n}{t} p^t (1-p)^{n-t}$$

$$\beta_{both}(p) = \mathbb{P}_p\left(\sum_{i=1}^n X_i < c_1 \text{ or } \sum_{i=1}^n X_i > c_2\right)$$

$$= \mathbb{P}_p\left(\sum_{i=1}^n X_i < c_1\right) + \mathbb{P}_p\left(\sum_{i=1}^n X_i > c_2\right)$$

$$= F_{n,p}\left(\lceil c_1 - 1\rceil\right) + 1 - F_{n,p}\left(\lfloor c_2 \rfloor\right) \quad \left[F_{n,p} \text{ is the CDF of Bin}(n,p)\right]$$

To have the same size α, we need

$$\beta_{left}\left(\tfrac{1}{4}\right) = \beta_{right}\left(\tfrac{1}{4}\right) = \beta_{both}\left(\tfrac{1}{4}\right) = \alpha$$

$$\Leftrightarrow F_{n,\frac{1}{4}}\left(\lceil c_{left} - 1\rceil\right) = 1 - F_{n,\frac{1}{4}}\left(\lfloor c_{right}\rfloor\right) = F_{n,\frac{1}{4}}\left(\lceil c_1 - 1\rceil\right) + 1 - F_{n,\frac{1}{4}}\left(\lfloor c_2\rfloor\right) = \alpha$$

Clearly, we should take C_{left}, C_{right} & (C_1, C_2) to be integers.

We should also take $C_{left} - 1 = b_{\alpha; n, \frac{1}{4}}$
$C_{right} = b_{1-\alpha; n, \frac{1}{4}}$

$$\left.\begin{aligned} C_1 - 1 &= b_{\alpha_1; n, \frac{1}{4}} \\ C_2 &= b_{1-\alpha_2; n, \frac{1}{4}} \end{aligned}\right\} \text{ with } \alpha_1 + \alpha_2 = \alpha. \text{ In particular, } \alpha_1 = \alpha_2 = \frac{\alpha}{2}.$$

Here, $b_{\alpha; n, \frac{1}{4}}$ is the lower α-quantile of $Bin(n, \frac{1}{4})$

i.e., $b_{\alpha; n, \frac{1}{4}}$ satisfies $\mathbb{P}\left(Bin(n, \frac{1}{4}) \le b_{\alpha; n, \frac{1}{4}}\right) = \alpha$

Note: In this discrete case, we cannot have a test with size α for every possible α. We can only have size α tests with α being all the different values that the cdf of $Bin(n, \frac{1}{4})$ takes.

$$C_{left} = b_{\alpha; n, \frac{1}{4}} + 1 \qquad \left| \qquad \left.\begin{aligned} C_1 &= b_{\alpha_1; n, \frac{1}{4}} \\ C_2 &= b_{1-\alpha_2; n, \frac{1}{4}} \end{aligned}\right\} \begin{aligned} &\alpha_1 + \alpha_2 = \alpha \\ &\text{e.g. } \alpha_1 = \alpha_2 = \frac{\alpha}{2}. \end{aligned}\right.$$
$$C_{right} = b_{1-\alpha; n, \frac{1}{4}}$$

From the power curves, observe that

φ_{left} is most appropriate against $\mathcal{H}_{left}$
φ_{right} is most appropriate against $\mathcal{H}_{right}$
φ_{both} is most appropriate against $\mathcal{H}_{both}$

Note: You may also study the effect of the choice of (α_1, α_2) on the power of φ_{both}. Take different choices of $\alpha_1 \in (0, \alpha)$ and set $\alpha_2 = \alpha - \alpha_1$. Draw the power curves of these tests on the same graph. What do you see?

(b) $X_1, \ldots, X_n \overset{iid}{\sim} \text{Poisson}(\lambda)$

$$\Rightarrow \sum_{i=1}^{n} X_i \sim \text{Poisson}(n\lambda)$$

Power functions : $\beta_{left}(\lambda) = \mathbb{P}_\lambda\left(\sum_{i=1}^{n} X_i < c_{left}\right)$

$$= \mathbb{P}_\lambda\left(\text{Poi}(n\lambda) < c_{left}\right)$$

$$= F_{n,\lambda}(\lceil c_{left} - 1 \rceil) \quad \left[\begin{array}{l} F_{n,\lambda}(\cdot) \text{ is the CDF of} \\ \text{Poi}(n\lambda) \end{array}\right]$$

$$= \sum_{t=0}^{\lceil c_{left}-1 \rceil} e^{-n\lambda} \frac{(n\lambda)^t}{t!}$$

$$\beta_{right}(\lambda) = \mathbb{P}_\lambda\left(\sum_{i=1}^{n} X_i > c_{right}\right)$$

$$= 1 - \mathbb{P}_\lambda\left(\sum_{i=1}^{n} X_i \le c_{right}\right)$$

$$= 1 - \mathbb{P}_\lambda\left(\text{Poi}(n\lambda) \le c_{right}\right)$$

$$= 1 - F_{n,\lambda}(\lfloor c_{right} \rfloor)$$

$$= 1 - \sum_{t=0}^{\lfloor c_{right} \rfloor} e^{-n\lambda} \frac{(n\lambda)^t}{t!} = \sum_{t=\lfloor c_{right} \rfloor + 1}^{\infty} e^{-n\lambda} \frac{(n\lambda)^t}{t!}$$

$$\beta_{both}(\lambda) = \mathbb{P}_\lambda\left(\text{Poi}(n\lambda) < c_1\right) + \mathbb{P}_\lambda\left(\text{Poi}(n\lambda) > c_2\right)$$

$$= F_{n,\lambda}(\lceil c_1 - 1 \rceil) + 1 - F_{n,\lambda}(\lfloor c_2 \rfloor)$$

To have the same size α, we need

$$\beta_{left}(2) = \beta_{right}(2) = \beta_{both}(2) = \alpha$$

$$\Leftrightarrow F_{n,\lambda}(\lceil c_{left} - 1 \rceil) = 1 - F_{n,\lambda}(\lfloor c_{right} \rfloor) = F_{n,\lambda}(\lceil c_1 - 1 \rceil) + 1 - F_{n,\lambda}(\lfloor c_2 \rfloor)$$

$$= \alpha$$

As in the previous case, we should take c_{left}, c_{right}, c_1 & c_2 to be integers ($\because$ the dist. of $\sum_{i=1}^{n} X_i$ is discrete)
Also, we should these values as follows.

$$\left.\begin{array}{l} c_{left} = P_{\alpha;n,\lambda}+1 \\ c_{right} = P_{1-\alpha;n,\lambda} \\ \left.\begin{array}{l} c_1 = P_{\alpha_1;n,\lambda}+1 \\ c_2 = P_{1-\alpha_2;n,\lambda} \end{array}\right\} \text{ with } \alpha_1+\alpha_2=\alpha \end{array}\right]$$

$P_{\alpha;n,\lambda}$ is the lower α-quantile of Poisson(n,λ) dist.

From the plots of the power functions, observe that

φ_{left} is most appropriate against $\mathcal{H}_{left}$
φ_{right} is most appropriate against $\mathcal{H}_{right}$
φ_{both} is most appropriate against $\mathcal{H}_{both}$

(c) $X_1, \ldots, X_n \overset{iid}{\sim}$ Uniform$(0,\theta)$

Power function:

$$\beta_{left}(\theta) = \mathbb{P}_\theta(X_1 < c_{left}) = \mathbb{P}_\theta\left(U(0,\theta) < c_{left}\right)$$
$$= \text{CDF of Unif}(0,\theta) \text{ at } c_{left} = \frac{c_{left}}{\theta} \quad \left[\because X_1 \text{ is conts.}\right]$$

$$\beta_{right}(\theta) = \mathbb{P}_\theta(X_1 > c_{right}) = 1 - \mathbb{P}_\theta(X_1 \le c_{right})$$
$$= 1 - \text{CDF of } U(0,\theta) \text{ at } c_{right} = 1 - \frac{c_{right}}{\theta}$$

$$\beta_{both}(\theta) = \mathbb{P}_\theta(X_1 < c_1) + \mathbb{P}_\theta(X_1 > c_2) = \frac{c_1}{\theta} + 1 - \frac{c_2}{\theta}$$

To have the same size α, we need

$$\beta_{left}(1) = \beta_{right}(1) = \beta_{both}(1) = \alpha$$
$$\Leftrightarrow c_{left} = 1 - c_{right} = c_1 + 1 - c_2 = \alpha$$

$$\left.\begin{array}{l} c_{left} = \alpha \\ c_{right} = 1-\alpha \\ c_2 - c_1 = 1-\alpha \end{array}\right| \begin{array}{l} c_1 = \alpha_1,\ c_2 = 1-\alpha_2 \\ \text{with } \alpha_1+\alpha_2=\alpha \end{array}$$

Notes: • The lower α-quantile of $U(0,1)$ is α

So, the cut-off points are in line with our earlier derivations in (a) & (b).

• Since the dist. of X_1 is continuous, we can have tests of any size.

(d) $X_1, \dots, X_n \overset{iid}{\sim} N(\mu, 1) \Rightarrow X_1 + X_2 \sim N(2\mu, 2) \Rightarrow \frac{X_1+X_2-2\mu}{\sqrt{2}} \sim N(0,1)$

$$\beta_{left}(\mu) = \mathbb{P}_\mu(X_1 + X_2 < c_{left})$$

$$= \mathbb{P}_\mu\Big(\underbrace{\frac{X_1+X_2-2\mu}{\sqrt{2}}}_{\sim N(0,1)} < \frac{c_{left}-2\mu}{\sqrt{2}}\Big)$$

$$= \mathbb{P}_\mu\Big(N(0,1) < \frac{c_{left}-2\mu}{\sqrt{2}}\Big) = \Phi\Big(\frac{c_{left}-2\mu}{\sqrt{2}}\Big)$$

$$\beta_{right}(\mu) = \mathbb{P}_\mu(X_1 + X_2 > c_{right})$$

$$= 1 - \mathbb{P}_\mu(X_1 + X_2 \le c_{right})$$

$$= 1 - \Phi\Big(\frac{c_{right}-2\mu}{\sqrt{2}}\Big) = \Phi\Big(\frac{2\mu - c_{right}}{\sqrt{2}}\Big)$$

$$\beta_{both}(\mu) = \mathbb{P}_\mu(X_1 + X_2 < c_1) + \mathbb{P}_\mu(X_1 + X_2 > c_2)$$

$$= \Phi\Big(\frac{c_1-2\mu}{\sqrt{2}}\Big) + 1 - \Phi\Big(\frac{c_2-2\mu}{\sqrt{2}}\Big)$$

To have the same size α, we need

$$\beta_{left}(5) = \beta_{right}(5) = \beta_{both}(5) = \alpha$$

$$\Leftrightarrow \Phi\Big(\frac{c_{left}-10}{\sqrt{2}}\Big) = 1 - \Phi\Big(\frac{c_{right}-10}{\sqrt{2}}\Big)$$

$$= \Phi\Big(\frac{c_1-10}{\sqrt{2}}\Big) + 1 - \Phi\Big(\frac{c_2-10}{\sqrt{2}}\Big) = \alpha$$

$$\Rightarrow c_{left} = 10 - \sqrt{2}\, z_\alpha, \quad c_{right} = 10 - \sqrt{2}\, z_{1-\alpha} = 10 + \sqrt{2}\, z_\alpha$$

$c_1 = 10 - \sqrt{2}\, z_{\alpha_1}$, $\quad c_2 = 10 + \sqrt{2}\, z_{\alpha_2}$ with $\alpha_1 + \alpha_2 = \alpha$

Eg. $\alpha_1 = \alpha_2 = \alpha/2$ $\quad$ [z_α is the upper α-quantile of $N(0,1)$ i.e., $1 - \Phi(z_\alpha) = \alpha$]

(e) $X_1, \ldots, X_n \overset{iid}{\sim} N(0, \sigma^2) \Rightarrow \sum_{i=1}^n X_i \sim N(0, n\sigma^2)$

$\Rightarrow \dfrac{\sum_{i=1}^n X_i}{\sqrt{n}\,\sigma} \sim N(0,1)$

$$\beta_{left}(\sigma) = \mathbb{P}_\sigma\left(\sum_{i=1}^n X_i < c_{left}\right)$$

$$= \mathbb{P}_\sigma\left(N(0, n\sigma^2) < c_{left}\right)$$

$$= \Phi_{0, n\sigma^2}(c_{left}) = \Phi\left(\frac{c_{left}}{\sqrt{n}\,\sigma}\right)$$

$\Phi_{\mu, \sigma^2}(\cdot)$ is the CDF of $N(0, \sigma^2)$

Verify that $\Phi_{\mu,\sigma^2}(t) = \Phi\left(\frac{t-\mu}{\sigma}\right)$ $\quad$ [$\Phi(\cdot) = \Phi_{0,1}(\cdot)$]

$$\beta_{right}(\sigma) = \mathbb{P}_\sigma\left(\sum_{i=1}^n X_i > c_{right}\right)$$

$$= 1 - \mathbb{P}_\sigma\left(\sum_{i=1}^n X_i \leq c_{right}\right)$$

$$= 1 - \Phi_{0, n\sigma^2}(c_{right}) = 1 - \Phi\left(\frac{c_{right}}{\sqrt{n}\,\sigma}\right)$$

$$\beta_{both}(\sigma) = \mathbb{P}_\sigma\left(\sum_{i=1}^n X_i < c_1\right) + \mathbb{P}_\sigma\left(\sum_{i=1}^n X_i > c_2\right)$$

$$= \Phi_{0, n\sigma^2}(c_1) + 1 - \Phi_{0, n\sigma^2}(c_2)$$

$$= \Phi\left(\frac{c_1}{\sqrt{n}\,\sigma}\right) + 1 - \Phi\left(\frac{c_2}{\sqrt{n}\,\sigma}\right)$$

To have the same size α:

$$\beta_{left}(10) = \beta_{right}(10) = \beta_{both}(10) = \alpha$$

$$\Leftrightarrow \Phi\left(\frac{c_{left}}{\sqrt{n}\,10}\right) = 1 - \Phi\left(\frac{c_{right}}{\sqrt{n}\,10}\right) = \Phi\left(\frac{c_1}{\sqrt{n}\,10}\right) + 1 - \Phi\left(\frac{c_2}{\sqrt{n}\,10}\right) = \alpha$$

$\Rightarrow$ $c_{left} = -10\sqrt{n}\, z_\alpha$

$c_{right} = -10\sqrt{n}\, z_{1-\alpha} = 10\sqrt{n}\, z_\alpha$

$\left| \begin{array}{l} c_1 = -10\sqrt{n}\, z_{\alpha_1} \\ c_2 = 10\sqrt{n}\, z_{\alpha_2} \end{array} \right]$ $\alpha_1 + \alpha_2 = \alpha$

From the plots, observe that φ_{left} & φ_{right} have the same power. This is because

$$\beta_{left}(\sigma) = \Phi\left(\frac{-10\sqrt{n}\,\tau_\alpha}{\sqrt{n}\,\sigma}\right) = \Phi\left(-\frac{10\,\tau_\alpha}{\sigma}\right)$$

$$\beta_{right}(\sigma) = 1 - \Phi\left(\frac{10\sqrt{n}\,\tau_\alpha}{\sqrt{n}\,\sigma}\right) = 1 - \Phi\left(\frac{10\,\tau_\alpha}{\sigma}\right) = \Phi\left(-\frac{10\,\tau_\alpha}{\sigma}\right)$$

Also,

$$\beta_{both}(\sigma) = \Phi\left(\frac{-10\sqrt{n}\,\tau_{\alpha/2}}{\sqrt{n}\,\sigma}\right) + 1 - \Phi\left(\frac{10\sqrt{n}\,\tau_{\alpha/2}}{\sqrt{n}\,\sigma}\right)$$

$$= \Phi\left(-\frac{10\,\tau_{\alpha/2}}{\sigma}\right) + 1 - \Phi\left(\frac{10\,\tau_{\alpha/2}}{\sigma}\right)$$

$$= 2\left\{1 - \Phi\left(\frac{10\,\tau_{\alpha/2}}{\sigma}\right)\right\}$$

You can check that none of these tests is uniformly more powerful than the other.

Also, none of them are suitable against $\mathcal{H}_{left}$.

This particular example shows that the choice of the test statistic plays a very crucial role in a test's performance.

<u>Exercise 2</u> $\quad X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, 1), \ \mu \in \mathbb{R}$

To test $H_0: \mu = 5$ vs. $H_a: \mu > 5$

$$\varphi_1(\underline{x}) = \mathbb{1}\{x_1 > c_1\}, \quad \varphi_2(\underline{x}) = \mathbb{1}\left\{\sum_{i=1}^{n} x_i > c_2\right\}$$

(a) $$\beta_1(\mu) = \mathbb{P}_\mu(X_1 > c_1) = \mathbb{P}_\mu(N(\mu,1) > c_1)$$

$$= 1 - \Phi_{\mu,1}(c_1) = 1 - \Phi(c_1 - \mu)$$

$$\beta_2(\mu) = \mathbb{P}_\mu\left(\sum_{i=1}^{n} X_i > c_2\right) = \mathbb{P}_\mu(N(n\mu, n) > c_2)$$

$$= 1 - \Phi_{n\mu,n}(c_2) = 1 - \Phi\left(\frac{c_2 - n\mu}{\sqrt{n}}\right)$$

(b) To have the same size, say α, we need

$$\beta_1(5) = \alpha = \beta_2(5)$$

$$\Rightarrow 1 - \Phi(c_1 - 5) = \alpha = 1 - \Phi\left(\frac{c_2 - 5n}{\sqrt{n}}\right)$$

$$\Rightarrow c_1 - 5 = z_\alpha \quad , \quad \frac{c_2 - 5n}{\sqrt{n}} = z_\alpha$$

[z_α is the upper α-quantile of $N(0,1)$. $1 - \Phi(z_\alpha) = \alpha$]

$$\Rightarrow c_1 = 5 + z_\alpha , \quad c_2 = 5n + \sqrt{n}\, z_\alpha$$

So, φ_1 & φ_2 become

$$\varphi_1(\underline{x}) = \mathbb{1}\{x_1 > 5 + z_\alpha\},$$

$$\varphi_2(\underline{x}) = \mathbb{1}\left\{\sum_{i=1}^n x_i > 5n + \sqrt{n}\, z_\alpha\right\}$$

$$\Leftrightarrow \mathbb{1}\left\{\bar{x} > 5 + \frac{z_\alpha}{\sqrt{n}}\right\}$$

(c) With the choice of c_1, c_2 as in (b), the power functions become

$$\beta_1(\mu) = 1 - \Phi(5 + z_\alpha - \mu) = 1 - \Phi(z_\alpha - (\mu - 5))$$

$$\beta_2(\mu) = 1 - \Phi\left(\frac{5n + \sqrt{n}\, z_\alpha - n\mu}{\sqrt{n}}\right)$$

$$= 1 - \Phi\left(z_\alpha - \sqrt{n}(\mu - 5)\right)$$

Notice that $n \geq 1$. Thus, for $\mu > 5$, $\sqrt{n}(\mu - 5) \geq \mu - 5$

The inequality is strict for $n \geq 2$.

$$\sqrt{n}(\mu - 5) \geq \mu - 5$$

$$\Rightarrow z_\alpha - \sqrt{n}(\mu - 5) \leq z_\alpha - (\mu - 5)$$

$$\Rightarrow \quad \Phi\left(\tau_\alpha - \sqrt{n}(\mu-5)\right) \leq \Phi\left(\tau_\alpha - (\mu-5)\right) \quad [\Phi \text{ is a cdf and hence non-decreasing}]$$

$$\Rightarrow 1-\Phi\left(\tau_\alpha - \sqrt{n}(\mu-5)\right) \geq 1-\Phi\left(\tau_\alpha - (\mu-5)\right)$$

$$\Rightarrow \beta_2(\mu) \geq \beta_1(\mu) \qquad \forall \mu > 5$$

The inequality is strict for $n \geq 2$.

φ_1, φ_2 are tests of the same size.

φ_2 has more power than φ_1 against $H_a: \mu > 5$.

So, φ_2 is more appropriate for testing $H_0: \mu = 5$ vs. $H_a: \mu > 5$.

<u>Exercise 3</u> $X_1, \ldots, X_n \overset{iid}{\sim} \text{Uniform}(0, \theta)$, $\theta > 0$.

To test $H_0: \theta = 5$ vs. $H_a: \theta > 5$.

$$\varphi_1(\underline{x}) = \mathbb{1}\{x_1 > c_1\}, \quad \varphi_2(\underline{x}) = \mathbb{1}\{x_{(n)} > c_2\}$$

(a) $\beta_1(\theta) = \mathbb{P}_\theta(X_1 > c_1)$

$$= 1 - \mathbb{P}_\theta(X_1 \leq c_1) = 1 - \frac{c_1}{\theta} \quad [\text{for } c_1 \in (0,\theta)]$$

$$\beta_2(\theta) = \mathbb{P}_\theta(X_{(n)} > c_2)$$

$$= 1 - \mathbb{P}_\theta(X_{(n)} \leq c_2) = 1 - \mathbb{P}_\theta(X_i \leq c_2 \ \forall i)$$

$$= 1 - \prod_{i=1}^n \mathbb{P}_\theta(X_i \leq c_2) = 1 - \{\mathbb{P}_\theta(X_1 \leq c_2)\}^n$$

$$= 1 - \left(\frac{c_2}{\theta}\right)^n.$$

(b) To have the same size, say α,

$$\beta_1(5) = \beta_2(5) = \alpha$$

$$\Rightarrow 1 - \frac{c_1}{5} = \alpha = 1 - \left(\frac{c_2}{5}\right)^n$$

$$\Rightarrow c_1 = 5(1-\alpha) \quad , \quad c_2 = 5(1-\alpha)^{1/n}$$

(c) With c_1 & c_2 as above,

$$\beta_1(\theta) = 1 - \frac{5(1-\alpha)}{\theta} \quad , \quad \beta_2(\theta) = 1 - \left(\frac{5(1-\alpha)^{1/n}}{\theta}\right)^n$$

$$= 1 - \frac{5}{\theta}(1-\alpha) \qquad\qquad = 1 - \left(\frac{5}{\theta}\right)^n (1-\alpha)$$

For $\theta > 5$, $\frac{5}{\theta} < 1 \Rightarrow \frac{5}{\theta} \geq \left(\frac{5}{\theta}\right)^n \quad \forall\, n \geq 1$

$$\frac{5}{\theta} > \left(\frac{5}{\theta}\right)^n \quad \forall\, n \geq 2$$

$$\therefore \frac{5}{\theta}(1-\alpha) \geq \left(\frac{5}{\theta}\right)^n (1-\alpha) \quad \forall\, n \geq 1, \theta > 5$$

$$\Rightarrow 1 - \frac{5}{\theta}(1-\alpha) \leq 1 - \left(\frac{5}{\theta}\right)^n (1-\alpha) \quad \forall\, n \geq 1, \theta > 5$$

$$\Rightarrow \beta_1(\theta) \leq \beta_2(\theta) \quad \forall\, \theta > 5$$

The inequality is strict for all $n \geq 2$.

So, φ_2 is more appropriate for testing $H_0: \theta = 5$ vs. $H_a: \theta > 5$.